When any transaction is initiated, any **node** can verify it on the network in a few simple steps:

1. First, the node will **check that the transaction is properly signed by the sender's private key**, which ensures that the transaction is legitimate and not tampered with.

2. Next, the node will **check that the sender has enough *funds*** to complete the transaction. This is done by examining the sender's balance on the blockchain ledger.

3. Finally, the node will also validate the transaction **inputs** and **outputs**, making sure that the inputs being spent in the transaction are not already spent in another transaction and that the outputs are not exceeding the total supply.

As we'll see, the combination of **public key cryptography** and the **UTXO** (Unspent Transaction Output) system is used in **Bitcoin** to verify the authenticity of transactions and prevent fraud without a central authority. **Public key cryptography** ensures secure communication and transfers of funds, while UTXO maintains a record of all the funds on the network and prevents double-spending.

UTXO, which stands for "Unspent Transaction Output," is simply a record of all the available funds in the network that have not yet been spent.

7.1 Tracking Your Coin's Journey

In ***Bitcoin***, transactions work like breaking a big bill into smaller bills and giving them to different people. The change you receive from a transaction is called an unspent output and can be used as an input for a new transaction. Outputs in ***Bitcoin*** transactions can either be **spent** or **unspent**, and unspent outputs are considered valuable because they can be used in new transactions.

- Think of it like using multiple gift cards to pay for a purchase. The gift cards from previous transactions act as inputs, and the change you receive is represented by a new gift card with the remaining amount. This is similar to how ***Bitcoin*** transactions work with UTXOs.

What are UTXOs?

The balance of a wallet is the sum of all a user's UTXOs. UTXOs are used to track the ownership of ***bitcoin*** in the network. When a transaction is made, it creates new UTXOs, and when a transaction is spent, it uses up existing UTXOs.

- UTXOs are like digital coins in the world of ***Bitcoin***. It is the change that you get back after you spend some ***bitcoin***.